

local-1777880479327

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Engineering report

Generated

Project ID	local-1777880479327
Domain	commercial_hvac
Engine	PDF Engine v2 (One-Shot)

TOTALS AT A GLANCE

MODULES	BOM ROWS	UNIT COST
5	33	£7,387.50
CEILING	HEADROOM	
None	N/A	

Contents

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1. Brief

Subject

Space heating and low-temp hot water provision for small commercial buildings.

Constraints

Unit Cost Ceiling	£0.00
Max Mass	-
Target Process	Air-to-water vapor compression refrigeration cycle utilizing high efficiency R290 refrigerant.
Target Material	Stainless steel (BPHE heat exchanger), R290 Propane, Sheet metal enclosure, Acoustic damping materials.
Tolerance	COP >= 3.5 @ A2/W35; Max Dimensions 1200Wx800Dx1400H mm; Max Noise <65dB @ 1m.
Quantity	500 units per year

Compliance Notes

MCS Certification required for UK grants. R290 requires BS EN 378 A3 flammability compliance. Hydro-split design recommended to keep R290 outdoors.

2. Regulatory posture

MCS 007

not-started

Product Certification Scheme Requirements: Heat Pumps

Specific requirements necessary for an air-source heat pump to receive MCS certification in the UK.

APPLICABILITY

Mandatory for all heat pumps targeted at UK small commercial/domestic government grants.

DESIGN IMPACT

Must prove COP testing via accredited lab and utilize ISO 9001 factory production controls.

EVIDENCE REQUIRED

MCS UKAS Laboratory Test Report

OWNER

Compliance Engineer

GAP ACTION

Engage BRE or equivalent testing facility for early MCS design review.

BS EN 14511

not-started

Air conditioners, liquid chilling packages and heat pumps for space heating

Establishes the test conditions and methods for rating efficiency and capacity.

APPLICABILITY

Mandatory to validate the 30kW capacity and COP ≥ 3.5 at the exact A2/W35 condition.

DESIGN IMPACT

Defines the exact enthalpy and psychrometric conditions the prototype must endure in the climate chamber.

EVIDENCE REQUIRED

Third-party EN 14511 performance certificate

OWNER

Thermodynamics Lead

GAP ACTION

Book environmental chamber lab time for prototype validation.

BS EN 378

not-started

Refrigerating systems and heat pumps. Safety and environmental requirements

Specifies safety guidelines for design, construction, and installation depending on refrigerant flammability.

APPLICABILITY

Critical due to the use of highly flammable R290 (Class A3).

DESIGN IMPACT

Mandates sealed outdoor unit circuit, ATEX compliant electrical enclosures inside chassis, and ventilation zones.

EVIDENCE REQUIRED

Charge limit calculations and risk assessment matrix

OWNER

Safety & Hardware Engineer

GAP ACTION

Calculate max permissible R290 charge volume and alter internal unit layout to separate electricals.

PED 2014/68/EU

not-started

Pressure Equipment Directive

Legislation governing the design, manufacture, and conformity of pressure equipment.

APPLICABILITY

Applies to the hermetically sealed R290 circuit including compressor, piping, and SS heat exchanger.

DESIGN IMPACT

Requires burst-pressure rated brazing/piping and safety relief valves calibrated to R290 high-side pressures.

EVIDENCE REQUIRED

Notified Body design examination and burst test data

OWNER

Mechanical/Piping Engineer

GAP ACTION

Source PED-certified heat exchangers and valves, draft piping thickness calcs.

BS EN 12102

not-started

Measurement of airborne noise emitted by air conditioners and heat pumps
Standard for determining sound power levels dynamically.

APPLICABILITY

Mandatory to verify the product meets the strict < 65dB at 1m requirement.

DESIGN IMPACT

Drives the requirement for an acoustic jacket on the scroll compressor and EC owl-wing axial fans.

EVIDENCE REQUIRED

Acoustic chamber test report (Sound Power & Sound Pressure)

OWNER

Acoustics/NVH Engineer

GAP ACTION

Review fan RPM profiles and perform NVH simulation on the 1200mm chassis enclosure.

3. Sizing optimisation

INFEASIBLE — design constraints not satisfied.

Envelope

Kind	generic_room
Dimensions	5000 × 5000 × 3000 mm
Floor area	25 m²

Module Dimensions

MODULE	W × D × H (MM)	AREA M²	MOUNT	SCALED BY
refrigeration_circuit	2236 × 2236 × 2400	5	floor	mass_proportional
air_extraction_system	2236 × 2236 × 2400	5	floor	mass_proportional
chassis_enclosure_system	2236 × 2236 × 2400	5	floor	mass_proportional
atex_control_module	2236 × 2236 × 2400	5	floor	mass_proportional
indoor_hydronic_module	2236 × 2236 × 2400	5	floor	mass_proportional

Conflicts

- Total module footprint (25.0m2) exceeds envelope budget (21.3m2)

Recommendations

- Consider a larger envelope
- Optimize module footprint

4.1. R290 Vapor-Compression Circuit

Lead: 12 weeks

Purpose

Executes the core thermodynamic cycle to extract and upgrade low-grade ambient heat into usable building warmth using R290.

Why it matters

It is the functional core of the heat pump. The performance of this module dictates whether the system can achieve the strict COP ≥ 3.5 target at A2/W35 conditions, while its structural integrity is mandatory for passing PED 2014/68/EU safety directives.

Description

This hermetically sealed thermodynamic module operates the essential heating cycle and contains the 2-4kg R290 propane charge. It compresses low-pressure, low-temperature vapor into a high-temperature gas, transfers the heat to a separate building water loop via the BPHE condenser, and throttles the liquified refrigerant back to the evaporator.

KEY PARTS

- Copeland Scroll Compressor (R290 specific)
- Stainless Steel Brazed Plate Heat Exchanger (BPHE)
- Electronic Expansion Valve (EEV)
- PED-certified copper pipework

FAILURE MODES

- Refrigerant leak at brazed joints
- Scroll compressor mechanical seizure
- BPHE catastrophic freezing or rupture

OPEN QUESTIONS

- PED burst-pressure requirements for the specific BPHE casing
- Exact optimum R290 charge volume per unit capacity

Cost Breakdown

PART #	NAME	TYPE	COST
refrigeration_circuit-001	Copeland Scroll Compressor (R290 specific)	COTS	£800.00
refrigeration_circuit-002	Stainless Steel Brazed Plate Heat Exchanger (BPHE)	COTS	£600.00
refrigeration_circuit-003	Electronic Expansion Valve (EEV)	COTS	£150.00
refrigeration_circuit-004	PED-certified copper pipework	COTS	£40.00
refrigeration_circuit-005	Compressor Vibration Isolation Grommets	COTS	£12.00
refrigeration_circuit-006	High/Low Pressure Transducers	COTS	£55.00
refrigeration_circuit-007	Elastomeric Pipe Thermal Insulation	COTS	£25.00
Module Total:			£1,682.00

Engineering Review

WARN

While thermally efficient, the use of a 2-4kg R290 charge presents severe flammability risks requiring ATEX-certified manufacturing environments and zero-tolerance leak detection.

RISK

2-4kg of R290 (propane) is a significant Class A3 highly flammable charge. Any leak during the charging process or testing poses an immediate explosion risk in the factory.

MANUFACTURABILITY

Manual brazing of PED-certified copper pipework for an A3 refrigerant introduces unacceptable human-error variability, leading to potential field leaks.

ENGINEERING

BPHE catastrophic freezing can occur if water flow drops during defrost cycles or low ambient operations.

RECOMMENDATIONS

- Upgrade the factory refrigerant charging zone to full ATEX zone compliance equipped with emergency exhaust systems.
- Implement a fully automated Helium mass spectrometry leak testing protocol in a vacuum chamber for all brazed assemblies prior to charging.
- Transition from manual gas brazing to automated induction brazing to ensure consistent penetration and hermetic sealing of joints.
- Add a primary water loop flow switch and freeze-protection algorithms to the unit's control board to prevent BPHE rupture.

4.2. Air Intake and Evaporator System

Lead: 10 weeks

Purpose

Transfers thermal energy from cold ambient air into the evaporating R290 refrigerant while maintaining strict acoustic limits.

Why it matters

This module directly controls the system's ability to satisfy the severe < 65dB at 1m acoustic limit (BS EN 12102) while extracting sufficient energy to deliver the 30kW capacity.

Description

Comprises the large frontal area heat exchanger (evaporator) with an optimized 2.0-2.5 mm fin pitch, paired with an oversized, electronically commutated (EC) axial fan featuring biomimetic swept blades. The fan pulls large volumes of outdoor air across the coil at low RPMs to boil the refrigerant without generating high fluid velocity noise.

KEY PARTS

- High-efficiency microchannel/fin-and-tube evaporator coil
- ebm-papst/Ziehl-Abegg EC Axial Fan
- Defrost drip pan and heating cable

FAILURE MODES

- Coil frost blockage restricting airflow
- EC motor or fan bearing failure
- Aerodynamic fan noise exceeding acoustic targets

OPEN QUESTIONS

- Dynamic sound power metrics at maximum required defrost fan RPM
- Real-world frost accumulation rates at nominal A2 conditions

Cost Breakdown

PART #	NAME	TYPE	COST
air_extraction_system-001	High-efficiency microchannel/fin-and-tube evap-COTS orator coil		£600.00
air_extraction_system-002	ebm-papst/Ziehl-Abegg EC Axial Fan	COTS	£350.00
air_extraction_system-003	Defrost drip pan and heating cable	COTS	£45.00
air_extraction_system-004	M8 Anti-Vibration Fan Mounting Kit	COTS	£15.00
air_extraction_system-005	Fan Safety Grille / Finger Guard	COTS	£22.00
Module Total:			£1,032.00

Engineering Review

WARN

While the acoustic and aerodynamic targets are well-founded, the combination of R290 refrigerant with a tight fin-pitch microchannel evaporator presents critical safety and cold-climate functional risks.

RISK

R290 is highly flammable (A3). The EC motor and defrost heating cables operate directly in the airstream and near the coil. In the event of a leak, these components pose severe ignition risks if not properly certified.

ENGINEERING

A fin pitch of 2.0-2.5 mm is overly dense for cold-ambient operation. Frost bridging will occur rapidly in humid conditions, restricting airflow and triggering excessive energy-intensive defrost cycles.

MATERIALS

Microchannel heat exchangers are excellent for condensers but notoriously problematic for cold-climate evaporators. Surface tension traps water within the micro-louvers during defrost, which refreezes and causes tube rupture.

MANUFACTURABILITY

Large 'oversized' axial fans with biomimetic blades are highly sensitive to dynamic balancing and housing concentricity.

RECOMMENDATIONS

- Switch evaporator architecture strictly to fin-and-tube with a hydrophilic coating rather than microchannel.
- Increase fin pitch to 3.0 mm minimum to extend the operational time between defrost cycles.
- Audit the Bill of Materials (BOM) to ensure the selected ebm-papst/Ziehl-Abegg fan and heating cable have specific A3/R290 ignition-proof ratings.
- Implement an active frost-thickness optical or pressure-drop sensor rather than relying on timed defrost algorithms to maximize acoustic and thermal efficiency.
- Review drip pan drainage angle and utilize a thermally-conductive, non-sparking element to prevent ice damming at the drain port.

4.3. Main Chassis and Acoustic Enclosure

Lead: 8 weeks

Purpose

Provides weatherproofing, acoustic damping, and the physical safety segregation necessary for the A3 flammable refrigerant.

Why it matters

Crucial for both BS EN 378 safety compliance (flammability segregation) and achieving the stringent acoustic requirements needed for UK deployment.

Description

The structural backbone of the outdoor unit, heavily engineered to conform to the compact 1200x800x1400mm dimensions. It incorporates specialized NVH damping materials to suppress compressor vibrations and includes a ventilated, compartmentalized sub-enclosure designed to prevent localized accumulation of leaking R290 gas.

KEY PARTS

- 1200x800x1400mm sheet metal cabinet
- Ventilated intrinsic safety sub-enclosure (for compressor)
- Mass-loaded vinyl acoustic dampening mats
- Hydrophobic acoustic foam layers

FAILURE MODES

- Sheet metal resonance amplifying compressor noise
- Corrosion of exterior panels
- Blockage of intrinsic safety ventilation pathways

OPEN QUESTIONS

- Specific resonant frequencies of the customized enclosure sheet metal
- Cost feasibility of sourcing the custom panels from overseas to meet £4,500 budget

Cost Breakdown

PART #	NAME	TYPE	COST
chassis_enclosure_system-001	1200x800x1400mm sheet metal cabinet	COTS	£40.00
chassis_enclosure_system-002	Ventilated intrinsic safety sub-enclosure (for compressor)	Custom	£800.00
chassis_enclosure_system-003	Mass-loaded vinyl acoustic dampening mats	COTS	£40.00
chassis_enclosure_system-004	Hydrophobic acoustic foam layers	COTS	£40.00
chassis_enclosure_system-005	Stainless Steel Fastener Kit (M6/M8 Bolts, Nuts, Washers)	COTS	£35.00
chassis_enclosure_system-006	Weather/Acoustic EPDM Panel Seals	COTS	£18.00
chassis_enclosure_system-007	A3 Flammable Gas Warning Label Kit	COTS	£5.00
Module Total:			£978.00

Engineering Review

WARN

The chassis design incorporates essential NVH and A3 safety features, but presents critical risks regarding R290 gas pooling and manufacturability challenges with acoustic material application.

RISK

Blockage of intrinsic safety ventilation pathways by snow, ice, or debris could lead to catastrophic R290 gas accumulation, as propane is heavier than air and sinks to the bottom.

MANUFACTURABILITY

Manual application of Mass-Loaded Vinyl (MLV) and hydrophobic acoustic foam is highly labor-intensive and prone to inconsistencies or delamination over time.

MATERIALS

The specification 'sheet metal cabinet' is too vague for an outdoor unit subjected to harsh environments, risking the noted corrosion failure mode.

ENGINEERING

Large flat panels on a 1200x1400mm surface are highly susceptible to 'oil-canning' and low-frequency compressor resonance.

RECOMMENDATIONS

- Conduct CFD (Computational Fluid Dynamics) simulations specifically modeling R290 leak dispersion with 50% restricted bottom vents to validate intrinsic safety.
- Update BOM to reflect specific anti-corrosion metal treatments and ATEX-compliant fasteners for the compressor sub-enclosure.
- Implement an automated or fixtured assembly process for MLV and foam installation to reduce takt time and ensure standardized acoustic performance.
- Introduce stamped stiffening features into the chassis panel tooling to mitigate structure-borne NVH passively, reducing the reliance on heavy MLV.

4.4. ATEX Electrical and Control Node

Lead: 14 weeks

Purpose

Manages system logic, power distribution, and dynamic component control safely within a potentially flammable environment.

Why it matters

Serves as the mandated safety barrier ensuring ignition sources (relays, arcs) never interact with A3 refrigerant, allowing the system to pass BS EN 378 regulations.

Description

Houses the control logic, inverter electronics, and power relays inside a strictly sealed, ATEX-rated enclosure separated from the primary refrigeration zone. It processes hydronic demands, orchestrates the speed of the fan and compressor, and serves as the ultimate failsafe to cut power in the event of an R290 leak detection.

KEY PARTS

- Inverter and Main Controller PCB
- ATEX-certified sealed electrical box
- Propane leak detection sensors
- Ignition-source isolated relays

FAILURE MODES

- Thermal runaway / overheating of the inverter
- Seal degradation compromising ATEX rating
- Sensor malfunction triggering false safety lockouts

OPEN QUESTIONS

- Capability to dissipate PCB heat from within a fully hermetic ATEX box
- Supply chain availability variations for certified spark-free contactors

Cost Breakdown

PART #	NAME	TYPE	COST
atex_control_module-001	Inverter and Main Controller PCB	COTS	£400.00
atex_control_module-002	ATEX-certified sealed electrical box	COTS	£15.00
atex_control_module-003	Propane leak detection sensors	COTS	£60.00
atex_control_module-004	Ignition-source isolated relays	COTS	£40.00
atex_control_module-005	ATEX Ex d/e Rated Cable Glands	COTS	£45.00
atex_control_module-006	Intrinsically Safe Sensor Wiring Harness	COTS	£60.00
atex_control_module-007	DIN Rail & PCB Nylon Standoffs	COTS	£8.00
Module Total:			£628.00

Engineering Review

WARN

While the safety architecture is sound in principle, placing a high-heat component like an inverter in a strictly sealed ATEX enclosure presents a critical thermal management contradiction that must be resolved.

ENGINEERING

Enclosing an inverter inside a strictly sealed ATEX box without active air exchange will lead to rapid thermal runaway and premature component failure.

MANUFACTURABILITY

Manual assembly is insufficient to guarantee ATEX seal integrity. Seal degradation or improper torque on fasteners during high-volume production is a major risk.

MATERIALS

Standard elastomer gaskets will suffer from compression set and thermal degradation over time, comprising the ATEX rating.

TOLERANCES

Mating surfaces of the enclosure must securely hold the gasket without pinching or allowing micro-gaps.

RISK

Propane leak sensors connected to the main control board pose a localized ignition risk if the wiring loops are not galvanically isolated.

RECOMMENDATIONS

- Redesign the enclosure as an Ex d (flameproof) die-cast aluminum housing with integrated thermal pathways and external heat sink fins.
- Integrate intrinsically safe (Ex i) barriers for all external sensor connections.
- Add 100% automated pressure decay leak testing to the assembly line layout.
- Introduce a hardware-based debounce and redundancy logic for the propane sensors to mitigate false safety lockouts.
- Define a strict torque sequence and specify captive fasteners for the enclosure lid to ensure even gasket compression.

4.5. Indoor Hydronic Module (Hydrobox)

Lead: 6 weeks

Purpose

Bridges the gap between the outdoor heat pump and the indoor building heating loop while preventing flammable gases from entering the habitable space.

Why it matters

It is the direct enabler for avoiding complex BS EN 378 indoor charge limits while providing the necessary user control and supplemental heating during defrosts.

Description

An indoor-mounted packaged assembly designed for hydronic splits. It pulls thermal energy indirectly from the outdoor BPHE condenser and distributes it throughout the building. By isolating the water loop components indoors, the design successfully prevents the entire R290 subsystem from entering constrained indoor spaces.

KEY PARTS

- Water circulation pump
- Expansion vessel
- Backup electrical immersion heater
- Indoor flow controller interface

FAILURE MODES

- Circulation pump mechanical failure
- Immersion heater electrical burnout
- Loss of fluid pressure in primary loop

OPEN QUESTIONS

- Optimum sizing for the expansion vessel given diverse commercial application volumes
- Specific pressure drop requirements for the connected external manifolds

Cost Breakdown

PART #	NAME	TYPE	COST
indoor_hydronic_module-001	Water circulation pump	COTS	£250.00
indoor_hydronic_module-002	Expansion vessel	COTS	£40.00
indoor_hydronic_module-003	Backup electrical immersion heater	COTS	£40.00
indoor_hydronic_module-004	Indoor flow controller interface	COTS	£180.00
indoor_hydronic_module-005	3-bar Pressure Relief Valve (PRV)	COTS	£25.00
indoor_hydronic_module-006	Hydronic Pipe Unions and EPDM Washers	COTS	£30.00
indoor_hydronic_module-007	Wall Mount Bracket & Anchors	COTS	£40.00
Module Total:			£605.00

Engineering Review

WARN

The Hydrobox concept is highly effective for isolating flammable R290 outside the building, but it requires strict safety interlocks for the immersion heater and robust freeze protection for the outdoor water lines.

RISK

Immersion heater electrical burnout due to extreme temperatures or 'dry-firing' during a loss of fluid pressure event.

ENGINEERING

Because the BPHE condenser is outdoors, the water/glycol loop running outside is highly susceptible to freezing during power outages or pump failures in winter.

MANUFACTURABILITY

High component density (pump, expansion vessel, heater, controls) in a singular indoor wall-mounted box can lead to extreme assembly difficulty and poor service access.

MATERIALS

Circulation pump volutes made of cast iron are prone to corrosion if system water is not properly treated, potentially seizing the impeller.

RECOMMENDATIONS

- Integrate a mechanical or ultrasonic flow switch inline with the backup heater to physically prevent dry-firing.
- Add a dedicated low-pressure transducer to shut down the pump and trigger a user alert if primary loop pressure drops below the minimum threshold.
- Configure the internal layout to ensure the expansion vessel's Schrader valve and the pump head are accessible directly from the front panel.
- Include clear documentation and commissioning software steps to verify required anti-freeze/glycol concentrations for the outdoor pipe run.

5. BOM master — 33 rows

PART #	NAME	MODULE	TYPE	MASS	COST
refrigeration_circuit-001	Copeland Scroll Compressor (R290 specific) varies	R290 Vapor-Compression Circuit	COTS	2.5kg	£800.00
refrigeration_circuit-002	Stainless Steel Brazed Plate Heat Exchanger (BPHE) varies	R290 Vapor-Compression Circuit	COTS	2.5kg	£600.00
refrigeration_circuit-003	Electronic Expansion Valve (EEV) varies	R290 Vapor-Compression Circuit	COTS	2.5kg	£150.00
refrigeration_circuit-004	PED-certified copper pipework varies	R290 Vapor-Compression Circuit	COTS	2.5kg	£40.00
air_extraction_system-001	High-efficiency microchannel/fin-and-tube evaporator coil varies	Air Intake and Evaporator System	COTS	3.333333333333335-kg	£600.00
air_extraction_system-002	ebm-papst/Ziehl-Abegg EC Axial Fan varies	Air Intake and Evaporator System	COTS	3.333333333333335-kg	£350.00
air_extraction_system-003	Defrost drip pan and heating cable varies	Air Intake and Evaporator System	COTS	3.333333333333335-kg	£45.00
chassis_enclosure_system-001	1200x800x1400mm sheet metal cabinet varies	Main Chassis and Acoustic Enclosure	COTS	2.5kg	£40.00
chassis_enclosure_system-002	Ventilated intrinsic safety sub-enclosure (for compressor) varies	Main Chassis and Acoustic Enclosure	Custom	2.5kg	£800.00
chassis_enclosure_system-003	Mass-loaded vinyl acoustic dampening mats varies	Main Chassis and Acoustic Enclosure	COTS	2.5kg	£40.00
chassis_enclosure_system-004	Hydrophobic acoustic foam layers varies	Main Chassis and Acoustic Enclosure	COTS	2.5kg	£40.00
atex_control_module-001	Inverter and Main Controller PCB FR4 varies	ATEX Electrical and Control Node	COTS	2.5kg	£400.00
atex_control_module-002	ATEX-certified sealed electrical box varies	ATEX Electrical and Control Node	COTS	2.5kg	£15.00
atex_control_module-003	Propane leak detection sensors varies	ATEX Electrical and Control Node	COTS	2.5kg	£60.00
atex_control_module-004	Ignition-source isolated relays varies	ATEX Electrical and Control Node	COTS	2.5kg	£40.00
indoor_hydraulic_module-001	Water circulation pump varies	Indoor Hydronic Module (Hydrobox)	COTS	2.5kg	£250.00
indoor_hydraulic_module-002	Expansion vessel varies	Indoor Hydronic Module (Hydrobox)	COTS	2.5kg	£40.00
indoor_hydraulic_module-003	Backup electrical immersion heater varies	Indoor Hydronic Module (Hydrobox)	COTS	2.5kg	£40.00
indoor_hydraulic_module-004	Indoor flow controller interface varies	Indoor Hydronic Module (Hydrobox)	COTS	2.5kg	£180.00

PART #	NAME	MODULE	TYPE	MASS	COST
refrigeration_circuit-005	Compressor Vibration Isolation Grommets EPDM Rubber	R290 Vapor-Compression Circuit	COTS	0.3kg	£12.00
refrigeration_circuit-006	High/Low Pressure Transducers Brass/Stainless Steel	R290 Vapor-Compression Circuit	COTS	0.15kg	£55.00
refrigeration_circuit-007	Elastomeric Pipe Thermal Insulation NBR Foam	R290 Vapor-Compression Circuit	COTS	0.5kg	£25.00
air_extraction_system-004	M8 Anti-Vibration Fan Mounting Kit Galvanized Steel/Rubber	Air Intake and Evaporator System	COTS	0.4kg	£15.00
air_extraction_system-005	Fan Safety Grille / Finger Guard Powder-coated Steel	Air Intake and Evaporator System	COTS	1.2kg	£22.00
chassis_enclosure_system-005	Stainless Steel Fastener Kit (M6/M8 Bolts, Nuts, Washers) 304 Stainless Steel	Main Chassis and Acoustic Enclosure	COTS	1.8kg	£35.00
chassis_enclosure_system-006	Weather/Acoustic EPDM Panel Seals EPDM Rubber	Main Chassis and Acoustic Enclosure	COTS	1.2kg	£18.00
chassis_enclosure_system-007	A3 Flammable Gas Warning Label Kit UV-resistant Vinyl	Main Chassis and Acoustic Enclosure	COTS	0.05kg	£5.00
atex_control_module-005	ATEX Ex d/e Rated Cable Glands Nickel-plated Brass	ATEX Electrical and Control Node	COTS	0.4kg	£45.00
atex_control_module-006	Intrinsically Safe Sensor Wiring Harness Copper/PVC/Techflex	ATEX Electrical and Control Node	COTS	0.6kg	£60.00
atex_control_module-007	DIN Rail & PCB Nylon Standoffs Aluminum/Nylon	ATEX Electrical and Control Node	COTS	0.2kg	£8.00
indoor_hydraulic_module-005	3-bar Pressure Relief Valve (PRV) Brass	Indoor Hydronic Module (Hydrobox)	COTS	0.35kg	£25.00
indoor_hydraulic_module-006	Hydronic Pipe Unions and EPDM Washers Brass/EPDM	Indoor Hydronic Module (Hydrobox)	COTS	0.8kg	£30.00
indoor_hydraulic_module-007	Wall Mount Bracket & Anchors Galvanized Steel	Indoor Hydronic Module (Hydrobox)	COTS	2.5kg	£40.00

Total Unit Cost: £4,925.00

6. Cost waterfall

UNIT COST	CEILING	HEADROOM
£7,387.50	None	N/A

Per-Module Roll-up

MODULE	% OF UNIT	COST
Module refrigeration_circuit	34.2%	£2,523.00
Module air_extraction_system	21.0%	£1,548.00
Module chassis_enclosure_system	19.9%	£1,467.00
Module atex_control_module	12.8%	£942.00
Module indoor_hydronic_module	12.3%	£907.50

NRE Total	£3,000.00
Overhead Multiplier	1.5×

Grand Total (Unit + NRE)	£10,387.50
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7. Risks register

R290 Vapor-Compression Circuit

Refrigerant leak resulting in flammable gas pooling		S:5 × L:2 = 10
CONTROLS	MITIGATION	
None	Specify burst-pressure rated brazing, perform rigorous factory pressure tests, and mandate sealed segregation from electrical systems.	
Owner: Safety & Hardware Engineer		

Air Intake and Evaporator System

Acoustic generation exceeds 65dB at 1 meter during peak RPM		S:4 × L:3 = 12
CONTROLS	MITIGATION	
None	Oversize the required fan diameter and specify sweeping 'owl-wing' blade geometries to reduce necessary rotational speeds.	
Owner: Acoustics/NVH Engineer		

Main Chassis and Acoustic Enclosure

Vibrational resonance in panels creating acoustic violations		S:4 × L:3 = 12
CONTROLS	MITIGATION	
None	Conduct FEA modal analysis on the sheet metal CAD and isolate compressor mounts with specifically tuned elastomers.	
Owner: Acoustics/NVH Engineer		

ATEX Electrical and Control Node

Electrical arc igniting ambient leaked propane

S:5 × L:1 = 5

CONTROLS

None

MITIGATION

Guarantee IP65 or better sealing on the electrical node and mechanically segregate it from the compressor chamber.

Owner: Safety & Hardware Engineer

Indoor Hydronic Module (Hydrobox)

Primary outdoor water loop freezing during severe cold power outage

S:3 × L:3 = 9

CONTROLS

None

MITIGATION

Design auto-drain purge valves into the system or mandate glycol anti-freeze in the installation manual.

Owner: Mechanical/Piping Engineer

8. Supplier shortlist — 33 parts

Copeland Scroll Compressor (R290 specific) (refrigeration_circuit-001)

Stainless Steel Brazed Plate Heat Exchanger (BPHE) (refrigeration_circuit-002)

Electronic Expansion Valve (EEV) (refrigeration_circuit-003)

PED-certified copper pipework (refrigeration_circuit-004)

High-efficiency microchannel/fin-and-tube evaporator coil (air_extraction_system-001)

ebm-papst/Ziehl-Abegg EC Axial Fan (air_extraction_system-002)

Defrost drip pan and heating cable (air_extraction_system-003)

1200x800x1400mm sheet metal cabinet (chassis_enclosure_system-001)

Ventilated intrinsic safety sub-enclosure (for compressor) (chassis_enclosure_system-002)

Mass-loaded vinyl acoustic dampening mats (chassis_enclosure_system-003)

Hydrophobic acoustic foam layers (chassis_enclosure_system-004)

Inverter and Main Controller PCB (atex_control_module-001)

ATEX-certified sealed electrical box (atex_control_module-002)

Propane leak detection sensors (atex_control_module-003)

Ignition-source isolated relays (atex_control_module-004)

Water circulation pump (indoor_hydraulic_module-001)

Expansion vessel (indoor_hydraulic_module-002)

Backup electrical immersion heater (indoor_hydraulic_module-003)

Indoor flow controller interface (indoor_hydraulic_module-004)

Compressor Vibration Isolation Grommets (gap-fill-1777880589487-0)

High/Low Pressure Transducers (gap-fill-1777880589487-1)

Elastomeric Pipe Thermal Insulation (gap-fill-1777880589487-2)

M8 Anti-Vibration Fan Mounting Kit (gap-fill-1777880589487-3)

Fan Safety Grille / Finger Guard (gap-fill-1777880589487-4)

Stainless Steel Fastener Kit (M6/M8 Bolts, Nuts, Washers) (gap-fill-1777880589487-5)

Weather/Acoustic EPDM Panel Seals (gap-fill-1777880589487-6)

A3 Flammable Gas Warning Label Kit (gap-fill-1777880589487-7)

ATEX Ex d/e Rated Cable Glands (gap-fill-1777880589487-8)

Intrinsically Safe Sensor Wiring Harness (gap-fill-1777880589487-9)

DIN Rail & PCB Nylon Standoffs (gap-fill-1777880589487-10)

3-bar Pressure Relief Valve (PRV) (gap-fill-1777880589487-11)

Hydronic Pipe Unions and EPDM Washers (gap-fill-1777880589487-12)

Wall Mount Bracket & Anchors (gap-fill-1777880589487-13)

9. Audit log

Pipeline execution log — generated by PDF Engine v2.

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